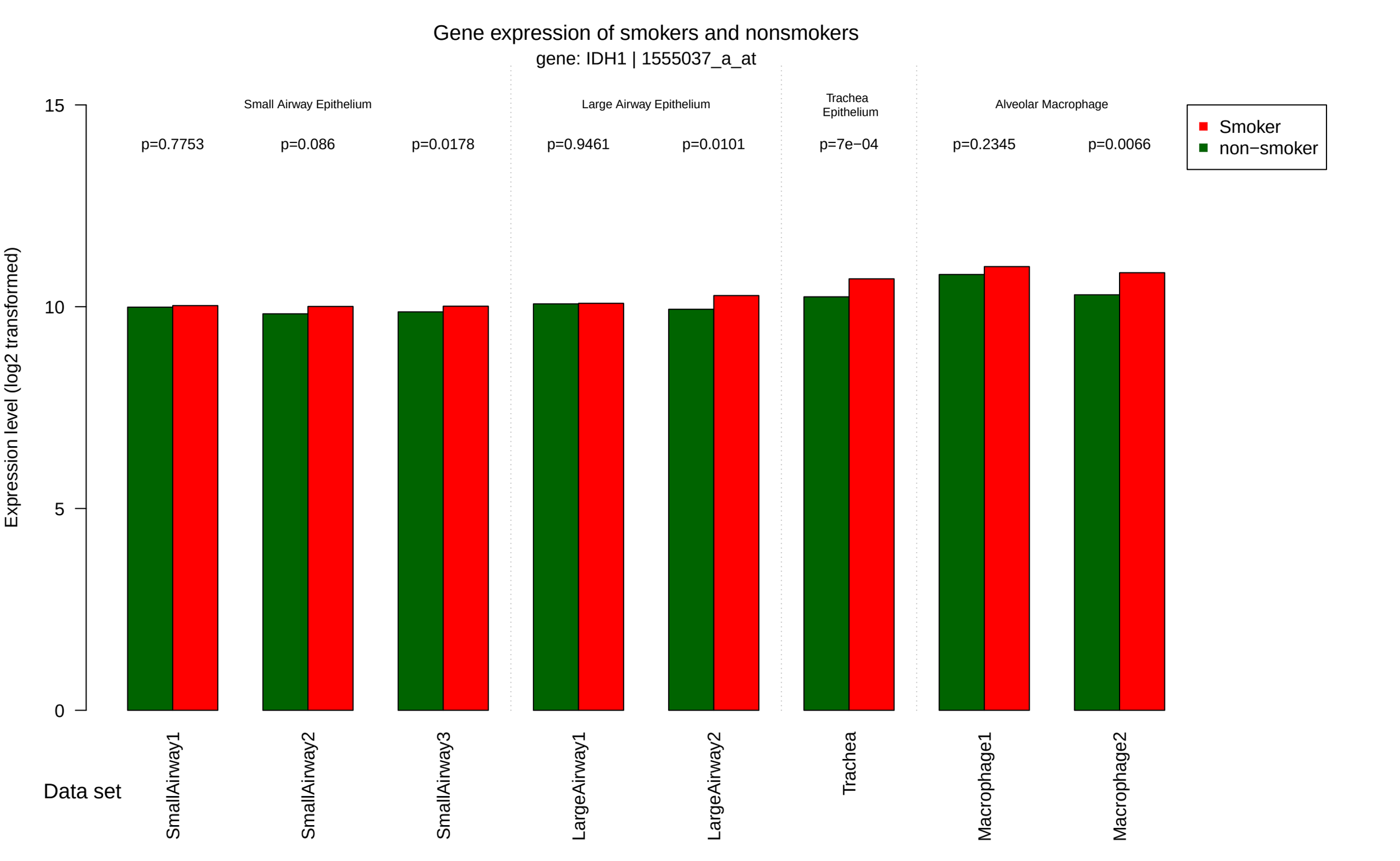
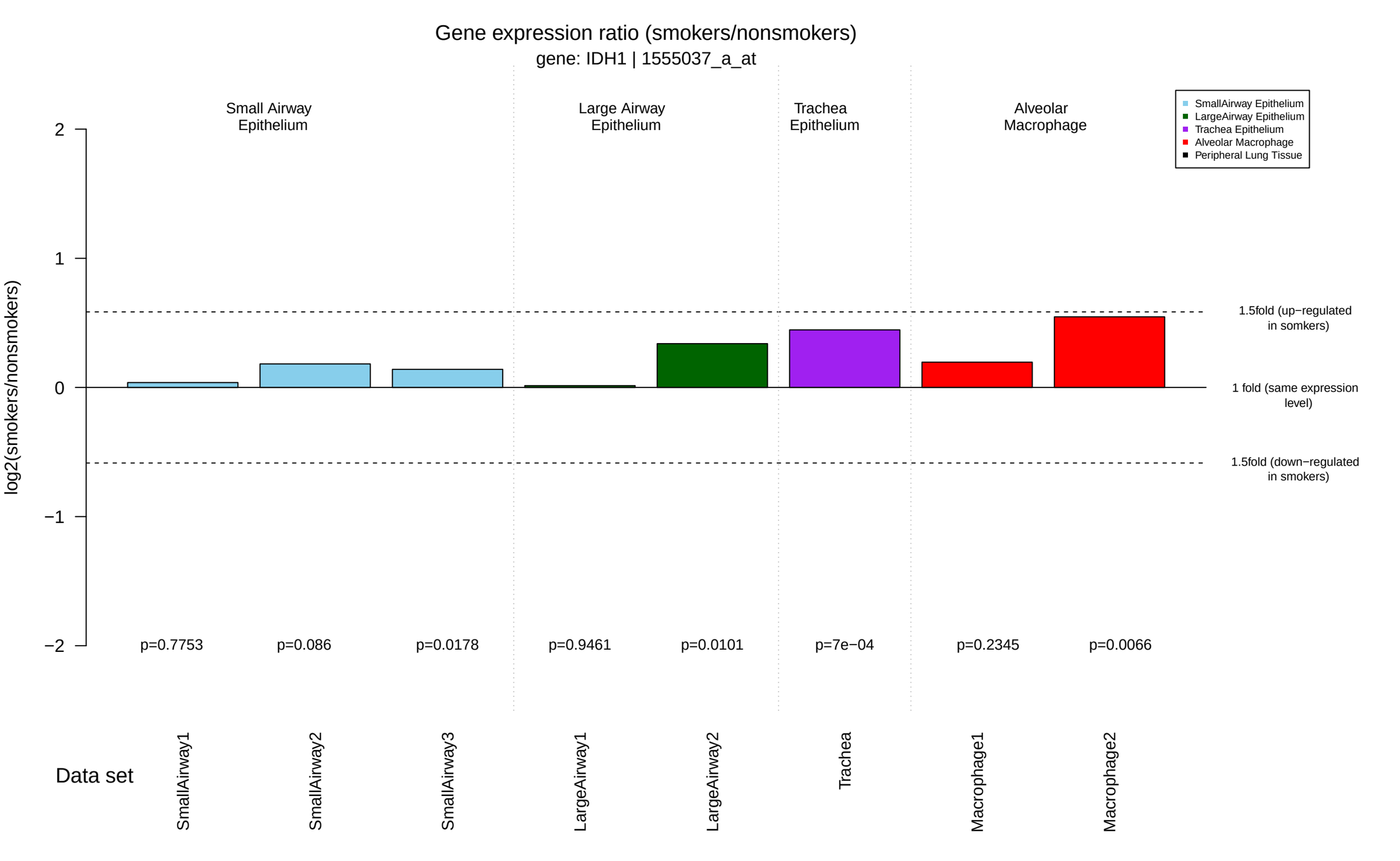
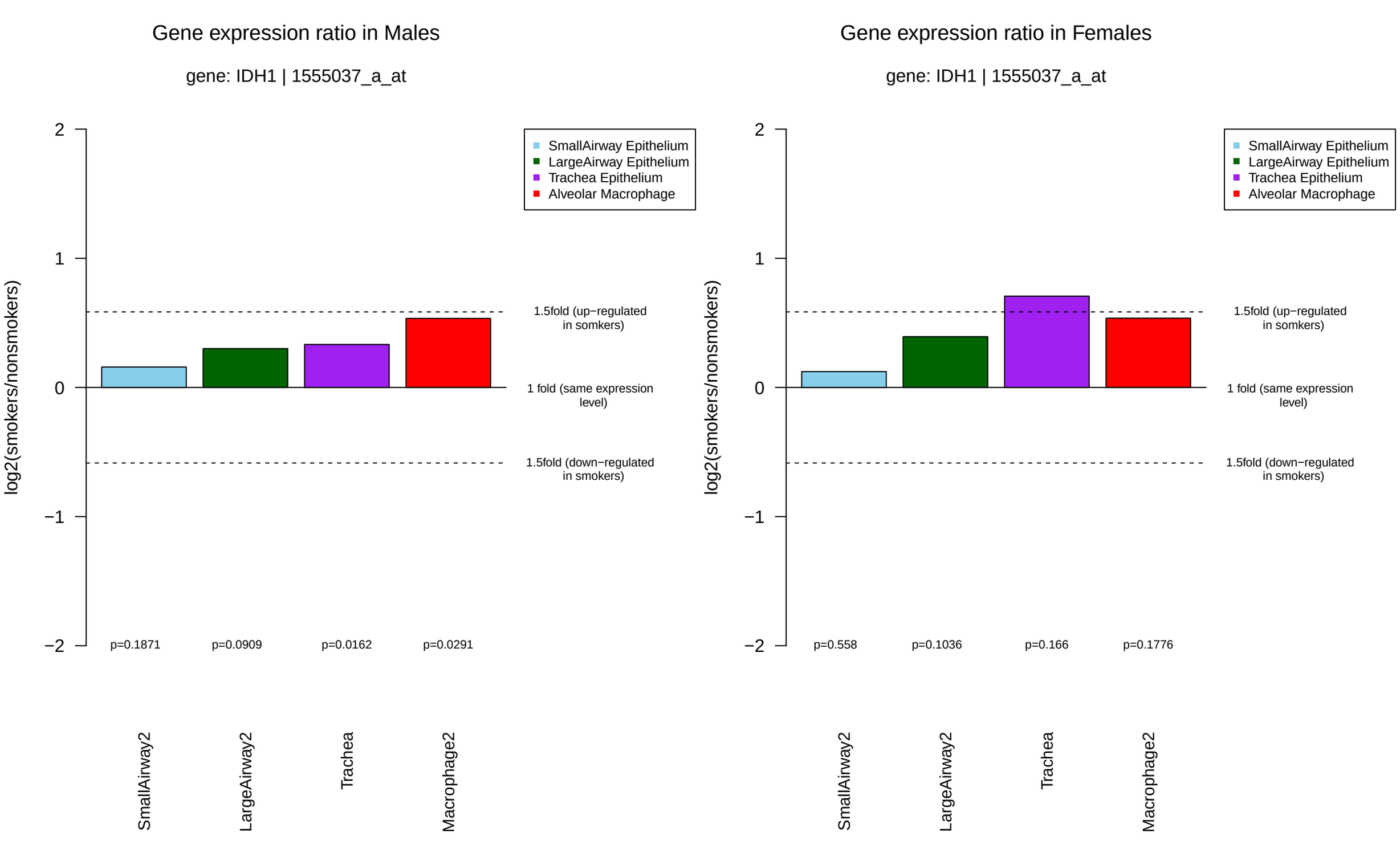
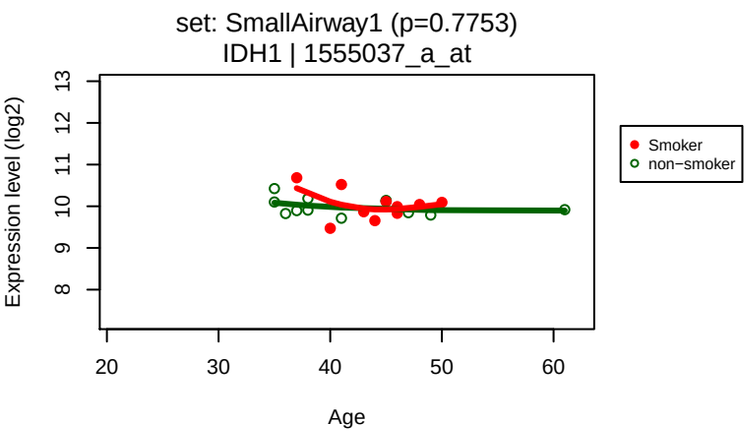
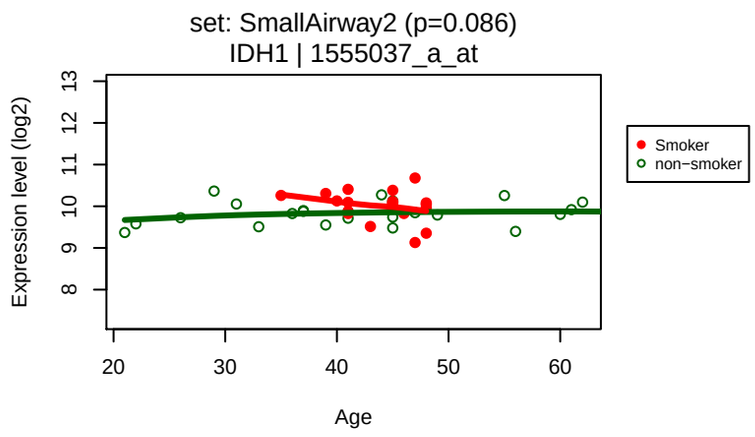
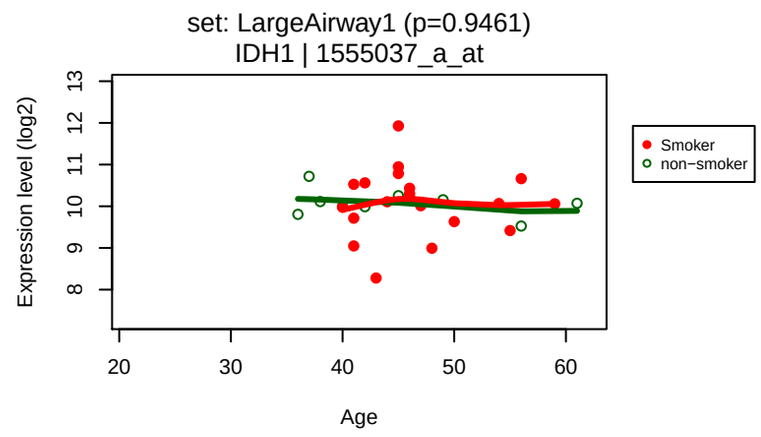

Gene expression of smokers and nonsmokers
gene: IDH1 | 1555037_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7753
p=0.086
p=0.0178
p=0.9461
p=0.0101
p=7e−04
p=0.2345
p=0.0066
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

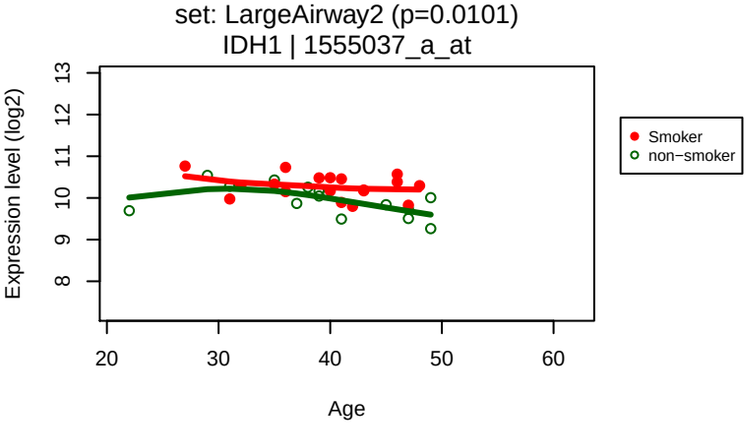

Gene expression ratio (smokers/nonsmokers)
gene: IDH1 | 1555037_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7753
p=0.086
p=0.0178
p=0.9461
p=0.0101
p=7e−04
p=0.2345
p=0.0066
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

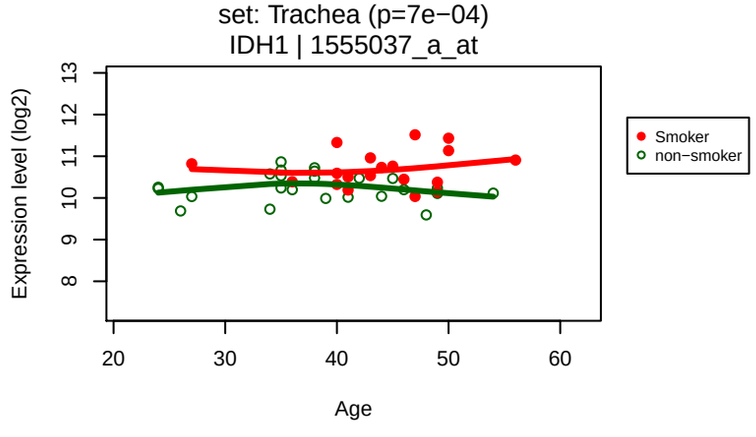

Gene expression ratio in Males
gene: IDH1 | 1555037_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1871
p=0.0909
p=0.0162
p=0.0291
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IDH1 | 1555037_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.558
p=0.1036
p=0.166
p=0.1776
SmallAirway2
LargeAirway2
Trachea
Macrophage2

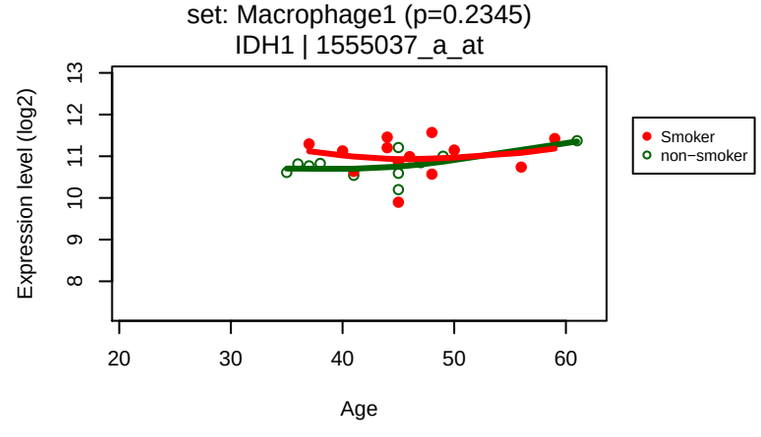

set: SmallAirway1 (p=0.7753)
IDH1 | 1555037_a_at
Expression level (log2)
Age
Smoker
non−smoker

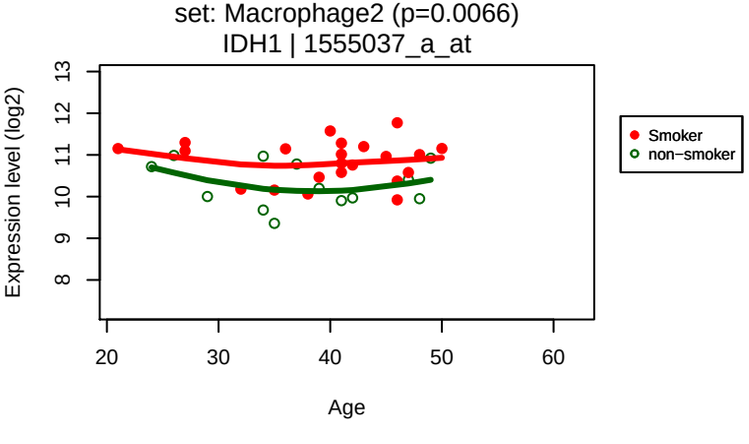

set: SmallAirway2 (p=0.086)
IDH1 | 1555037_a_at
Expression level (log2)
Age
Smoker
non−smoker

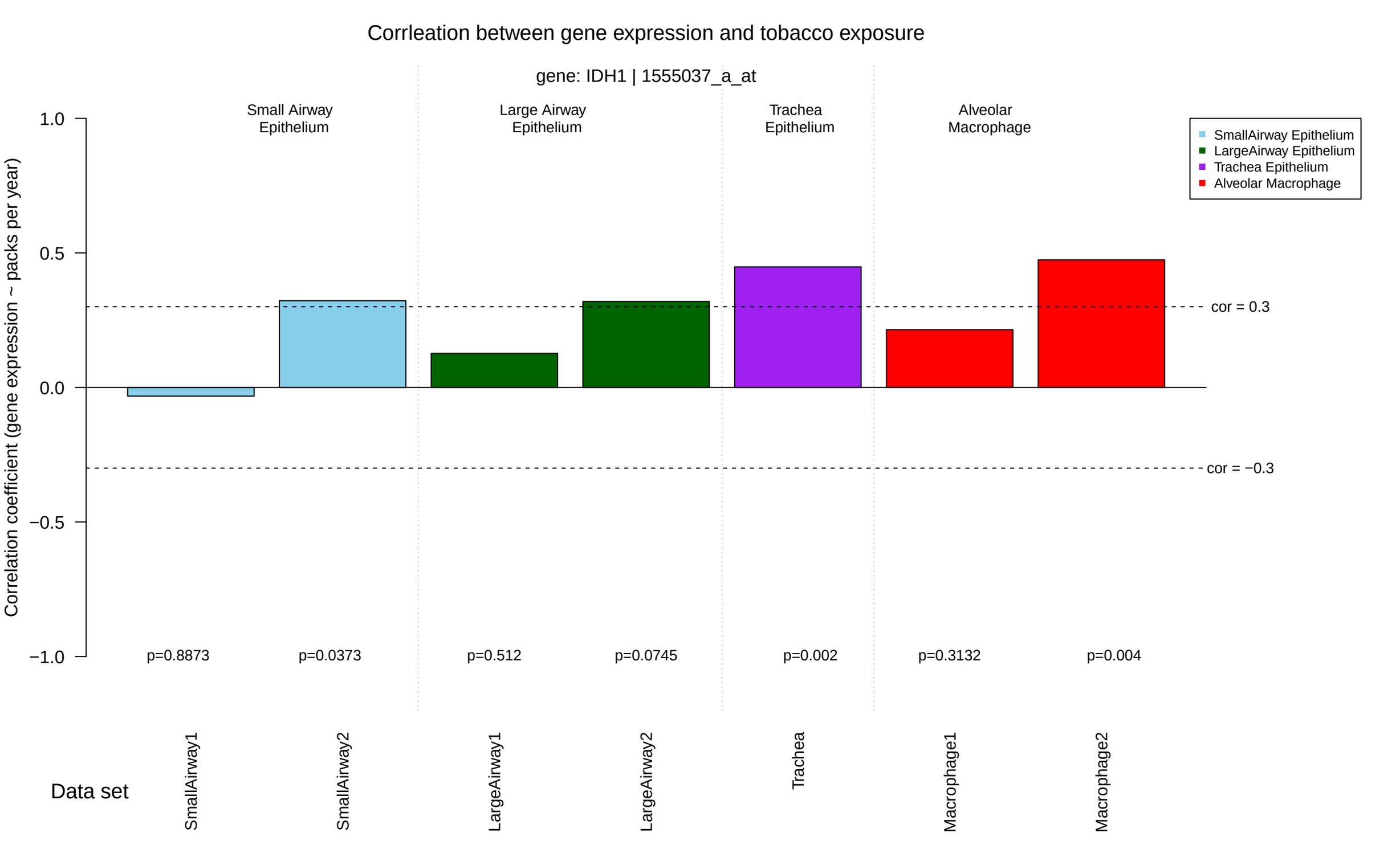

set: LargeAirway1 (p=0.9461)
IDH1 | 1555037_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0101)
IDH1 | 1555037_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=7e−04)
IDH1 | 1555037_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2345)
IDH1 | 1555037_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0066)
IDH1 | 1555037_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IDH1 | 1555037_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8873
p=0.0373
p=0.512
p=0.0745
p=0.002
p=0.3132
p=0.004
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2